

New European Concept of Single Staged Rocket to launch nano-satellites in Low Earth Orbit (LEO)

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Abstract - The possibility to launch small payloads into LEO by means of an electromagnetic railgun is the motivation of the present study. Here it is assumed that an electromagnetic railgun with barrel length of 180m is able to launch a 120 mm diameter projectile at a speed of 6 km/s. In order to the payload achieves orbital motion and to compensate the velocity losses due to gravity and aerodynamic drag, a rocket engine which supplies the required velocity difference is integrated into the projectile structure. The development of such propelled projectile requires key technologies that could also benefit other industrial sectors. The paper presents the main concept and design aspects to be considered for such propelled projectile with special emphasis on: flight mechanics, aero/thermodynamics, materials and propulsion characteristics.

The aerodynamic form of the projectile is optimized to achieve minimum drag and lower heat loads. Launch angles and trajectory optimization analysis are carried out by means of 3 degree of freedom simulations (3DOF). Furthermore, three concurrent propulsion systems are here considered for the rocket engine: solid, liquid and hybrid propellants type. The result of analysis pointed that the most promising propulsion systems are: (I) hybrid engine based on hydrogen peroxide/wax propellants and (II) liquid engine based on hydrogen peroxide/ hydrocarbon propellants. The structural mass is analyzed mostly on the basis of carbon fiber reinforced materials as well as classical aerospace metallic in small amount. Finally, the paper gives a critical mark for the technical feasibility of small rockets for such missions.

Key words: nano-satellite, one-stage-rocket, railgun, hybrid-rocket-engine, mass optimization

INTRODUCTION

EADS - Space Transportation is leading a study aimed to investigate the possibility to launch pico- and nano-satellites with scientific payloads into Low-Earth-Orbit (LEO), based on a combination of an electromagnetic railgun and a hypersonic projectile [2],[3]. Recent progress in the field [1] could allow the launch of a 42 kg projectile at a speed of 6000 m/s by means of a 180m length railgun. The projectile will be exposed to an initial acceleration of 10,000 g. In the frame of that investigation, the German Aerospace Center (DLR) is responsible

for the design of a one stage small hypersonic propelled projectile.

The main goal of the study is to determine the feasibility of such a projectile with emphasis in the flight mechanics, aero/thermodynamics and propulsion system concept and to identify the key-technologies associated to the railgun-launch system.

AEROTHERMODYNAMICS

To initiate the aerodynamic design loop, an initial shape and projectile dimensions are given: total length 1200 mm, body diameter 120mm, fin span 200mm. The nose cap has a contour according to the power law with an exponent $n = 0.62$ to guarantee minimum drag. A payload of 5kg is considered. The launch velocity from the railgun is 6km/s corresponding to a Mach number 17.7 at sea level.

Fig. 1 shows drag coefficients c_x as a function of Mach number Ma , calculated with a handbook methods. The hypersonic launch velocity is of great advantage since as the figure shows the drag coefficient c_x drastically decreases for Mach number $Ma \geq 4$. Fig. 2 shows a zoom of the calculated flight path for a propelled projectile in terms of time, for the first 20 seconds. The projectile passes the troposphere and achieves the stratosphere after about 4sec and it is already at 30km altitude after approx. 12sec. The calculation also shows that the projectile passes the densest atmosphere range after 20sec

The critical case for the dimensioning mechanical and thermals loads are short after the mission start, about 0.01sec flight time ($Ma = 17.69$). The flow around the projectile is very complex: viscous, turbulent and fully unsteady. Short after the start, real gas effects dominate the flow field. Furthermore, practical experiences or measurements on projectiles at high hypersonic velocities ($Ma > 15$) within dense atmosphere conditions close to sea level are not known.

Simulations of the 3D flow around the projectile configuration for such conditions are here done with the DLR Code TAU for the solution of the Reynolds-averaged Navier-Stokes equations (s. constant Mach number lines distribution in Fig. 4). The unstructured grid with 3.7 million points is generated with the CentaurSoft code and two times adapted (Fig. 3). The computed pressure distribution presented in Fig. 5 shows a strong pressure load at the tip of the nose cone of the projectile.

Assuming a 2 mm nose radius, the resulting absolute pressure value is $21.42 \cdot 10^6 \text{ N/m}^2$. However, short downstream the pressure drops to $3.2 \cdot 10^6 \text{ N/m}^2$ and further downstream to $0.3 \cdot 10^6 \text{ N/m}^2$.

The computed surface temperature distribution shows (Fig. 6) that the hottest locations are the tip of the nose cone, approx. 2900 K, the junction between fuselage and fins and at the fins leading edges, approx. 1700 K. The computations are done for steady conditions assuming that the thermal gradients acting on the projectile surface are almost the same as for unsteady conditions and scaling the absolute values with a corrected stagnation point value. The computed heat flux at the stagnation point is corrected by means an extrapolation factor attained through a similarity function from experiments in ISL¹ for lower mach numbers. If necessary the thermal protection of critical areas could be based on an ablation-, transpiration- or passive-cooling [5].

FLIGHT DYNAMIC

In order the payload achieves orbital motion and to compensate the velocity losses due to gravity and aerodynamic drag, a rocket engine which supplies the required velocity difference is integrated into the projectile structure. The trajectories, computed with the 3DOF DLR code TRAJECTORY-3D, are function of the aerodynamic and mass properties of the projectile, characteristics of the rocket engine and properties of the earth atmosphere and start conditions.

Mission profile

The propelled projectile is launched by the railgun at 6 km/s (Fig. 7). The projectile flies without propulsion till short before it reaches LEO where the angle of trajectory inclination becomes almost zero. There the projectile ignites its rocket motor which burns for approximately 10sec providing the projectile with the additional velocity to insert the payload in orbit.

Minimum projectile mass - trajectory optimization

To determine the minimum projectile total mass m_t which fulfill the mission and the required specific impulse I_{sp} and total impulse I_{tot} for the rocket engine, a number of trajectory computations are done accounting for different propellant performances, launch altitudes and elevation angles of the railgun muzzle θ . Here only those results achievable with today's or near future propellant technology are discussed. Also, for all calculated cases elevation angles of the railgun muzzle $\theta = 31^\circ$ were assumed, resulting from the energy balance between the gravitational and aerodynamic forces.

CASE 1: 35 kg is the minimum mass for a propelled projectile able to introduce 5 kg mass payload into LEO requiring also that the projectile empty mass m_k does not exceed 9 kg and the propellant mass m_f does not exceed 21kg (Fig. 9). Furthermore, it is required a liquid rocket motor with a minimum specific impulse in vacuum of

$I_{sp} = 450\text{sec}$ and total impulse $I_{tot} = 93,000\text{Nsec}$ in order to achieve a stable circular LEO orbit, a very high value today not achievable. Only few cryogenic and high toxic liquid and hybrid propellants have theoretical values close to 450sec. The velocity distribution along the projectile fly path is shown on Fig. 10.

CASE 2: Fig. 11 shows that a projectile with total mass $m_t = 41\text{kg}$, propellant mass $m_f = 29,4\text{kg}$ and a hybrid rocket motor with specific impulse in vacuum of $I_{sp} = 338\text{sec}$ and corresponding total impulse $I_{tot} = 97,600\text{Nsec}$ plenty fulfill the mission request to launch 5 kg payload in circular LEO. Fig. 12 show the velocity distribution as a function of range for the suborbital and orbital part of the trajectory. On the suborbital part of the trajectory short after the launch from the railgun can be seen a loss of velocity of the order of $\Delta V = 2500\text{m/s}$ due to aerodynamic drag.

A similar projectile equipped with a liquid rocket motor weights without propellants 2.5kg more. These few more kilograms however have consequences on the whole system. To launch 5 kg payload in circular LEO the projectile with a LRM (CASE 3) have a total mass $m_t = 46\text{kg}$, propellant mass $m_f = 32\text{kg}$, motor with $I_{sp} = 346\text{sec}$ (in vacuum) and corresponding $I_{tot} = 108,600\text{Nsec}$ (Fig. 13 and Fig. 14).

A small sounding rockets with non-cryogenic liquid propellants, which are common in use, have a lower specific impulse. This is the reason to consider CASE 4. Indeed, increasing 2kg of propellant mass with respect to CASE 3 result in a total impulse $I_{tot} = 111,700\text{Nsec}$, permitting decrease the specific impulse of the rocket engine to $I_{sp} = 335\text{sec}$ (Fig. 13).

PROPULSION SYSTEM

Three concurrent propulsions systems are considered for the rocket engine: solid, liquid and hybrid propellants type. The rocket engine properties, main dimensions and weights of the elements are estimated with the DLR code ENGINE-MOD. The most promising propulsion systems are: (I) a hybrid rocket motor (HRM) based on hydrogen peroxide/wax propellants (Fig. 15) and (II) a liquid rocket motor (LRM) based on hydrogen peroxide/hydrocarbon propellants. For both systems it is chosen (1) a simple pressure feeding system on the basis of helium as a working gas; (2) a simple control and monitoring system; (3) a combustion chamber with short burning time (approx. 10 sec) and a carbon based ablation cooling system; (4) an ablation cooled plug nozzle (aerospike). Also, it is proposed a high integration level of all components accomplished mainly by carbon fiber based composite technology.

Hybrid rocket motors (HRM) are in comparison with LRM based on liquid bipropellant $\text{H}_2\text{O}_2/\text{kerosene}$ lighter for more than 4 kg. Indeed, a HRM has not fuel tank since it is integrated in the combustion chamber. Also the injector, propellant valving, piping and tanks venting are simpler and lighter. Furthermore a HRM has lower number of components and lower production cost. The security aspects are clear advantage of the HRMs [4]. A

proposed projectile configuration propelled with HRM for LEO Payloads could have an overall length 2900mm and body diameter 120mm [3]. The HRM has a specific impulse in vacuum of $I_{sp}=318\text{sec}$ and total impulse of $I_{tot} = 97000\text{Nsec}$.

The above-required values of specific impulse can be reduced if the launch place lies not at sea level. The present study indicates a specific impulse reduction rate of about 15sec each 2000m altitude increase. Fig. 16 shows an artist impression of RAILGUN launching facility at high mountain.

CONCLUSION

It can be concluded that a launching system for small payloads into LEO composed of an electromagnetic railgun and a propelled projectile is today a realistic option. The present study showed that there are no marked critical problems or processes, which can not be controlled with today's technology.

ACKNOWLEDGEMENT

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NOMENCLATURE

d_s - body diameter
 I_{sp} - specific impulse
 I_{tot} - total impulse
 m_t - total projectile mass
 m_p - payload
 m_f - propellant mass
 m_k - construction mass
 θ - elevation angle of the railgun muzzle

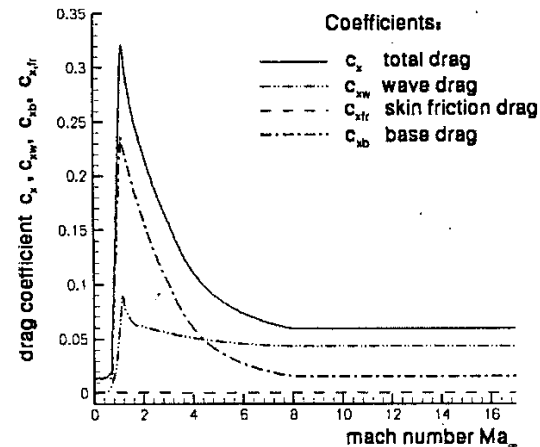


Fig. 1: Computed projectile aerodynamic drag as a function of Mach number

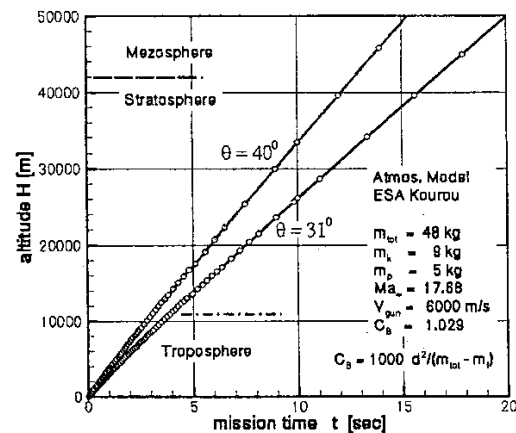


Fig. 2: Zoom of the calculated flight path for LEO Payload Propelled Projectile in term of time for the first 20sec

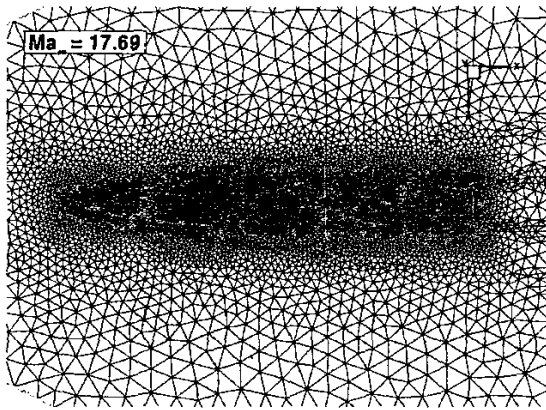


Fig. 3: Two time adapted unstructured N-S-grid with 3,7 mil. points

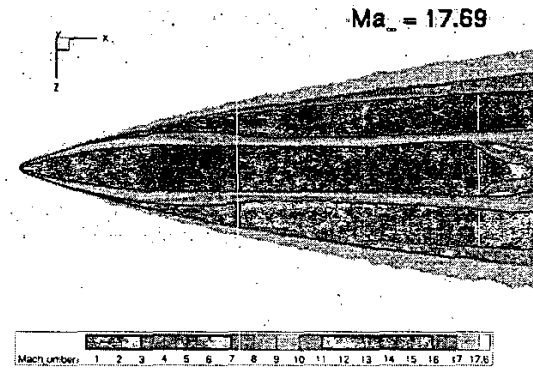


Fig. 4: Mach number contour distribution around the propelled projectile

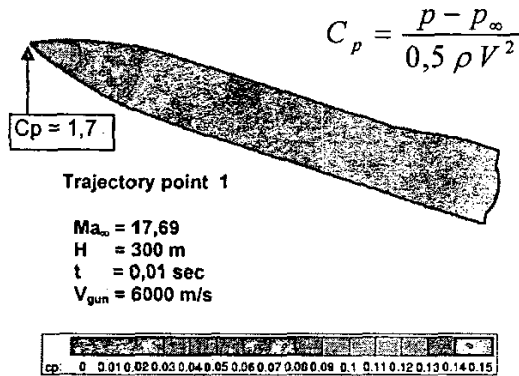


Fig. 5: Surface pressure distribution (initial propelled projectile configuration)

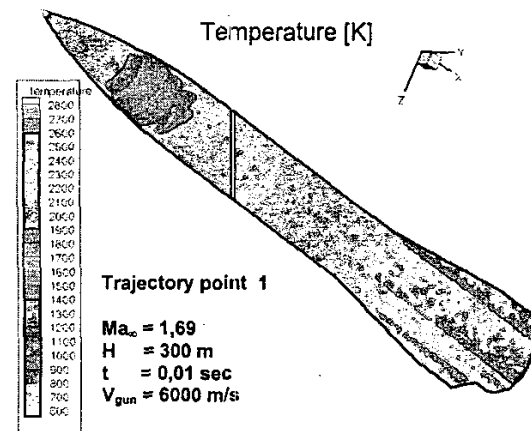


Fig. 6: Surface temperature distribution (initial propelled projectile configuration)

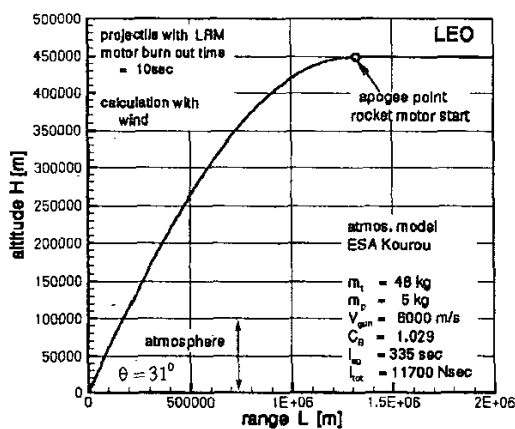


Fig. 7: Propelled projectile mission profile

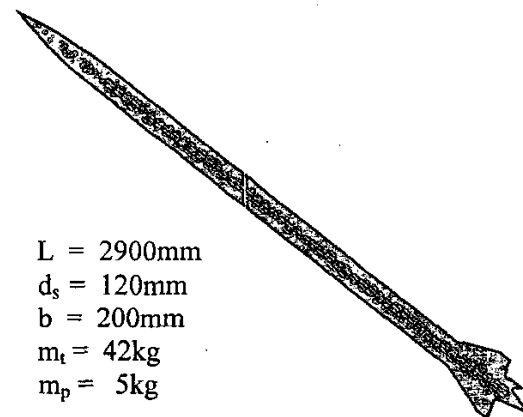


Fig. 8: Proposed propelled projectile configuration

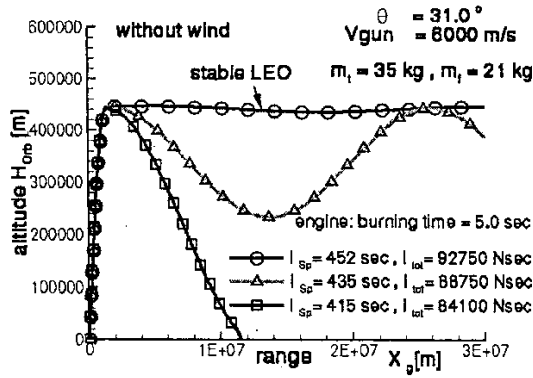


Fig. 9: Orbit insertion for CASE 1

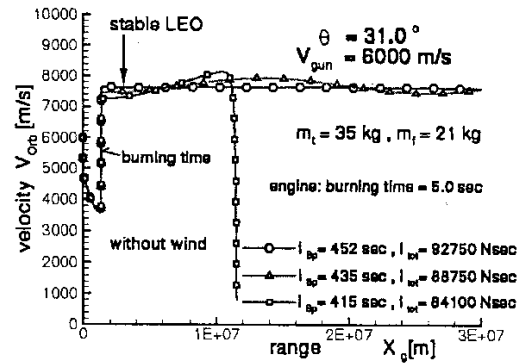


Fig. 10: Velocity distribution along the fly path for CASE 1

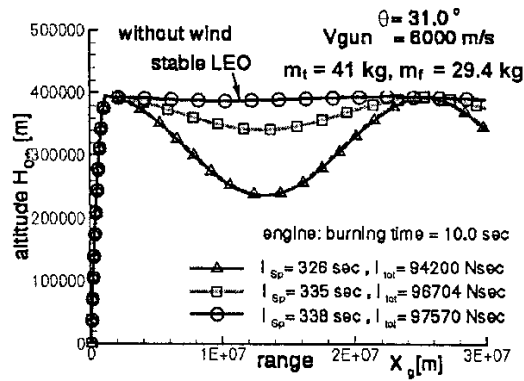


Fig. 11: Orbit insertion for CASE 2

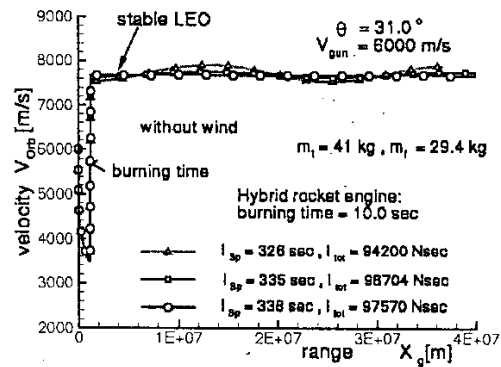


Fig. 12: Velocity distribution along the fly path for CASE 2

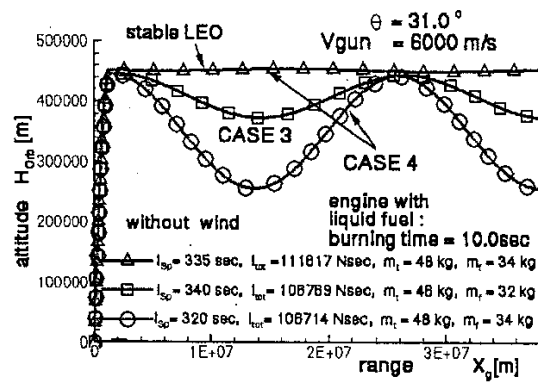


Fig. 13: Orbit insertion for CASE 3 and CASE 4

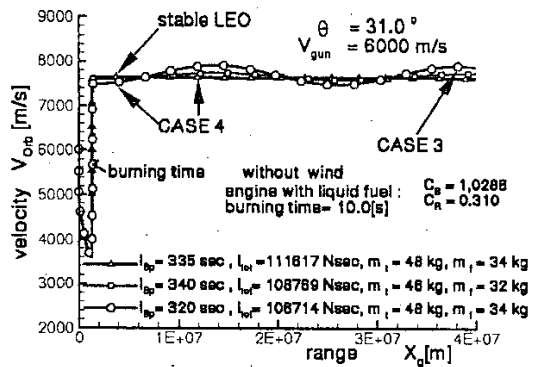


Fig. 14: Velocity distribution along the fly path for CASE 3 and CASE 4

CASE 1	$C_B = 1.028$	Ballistic coefficient
CASE 2	$C_B = 1.241$	
CASE 3	$C_B = 1.028$	
CASE 4	$C_B = 1.028$	

$$C_B = 10^3 \frac{d_s^2}{(m_t - m_r)}$$

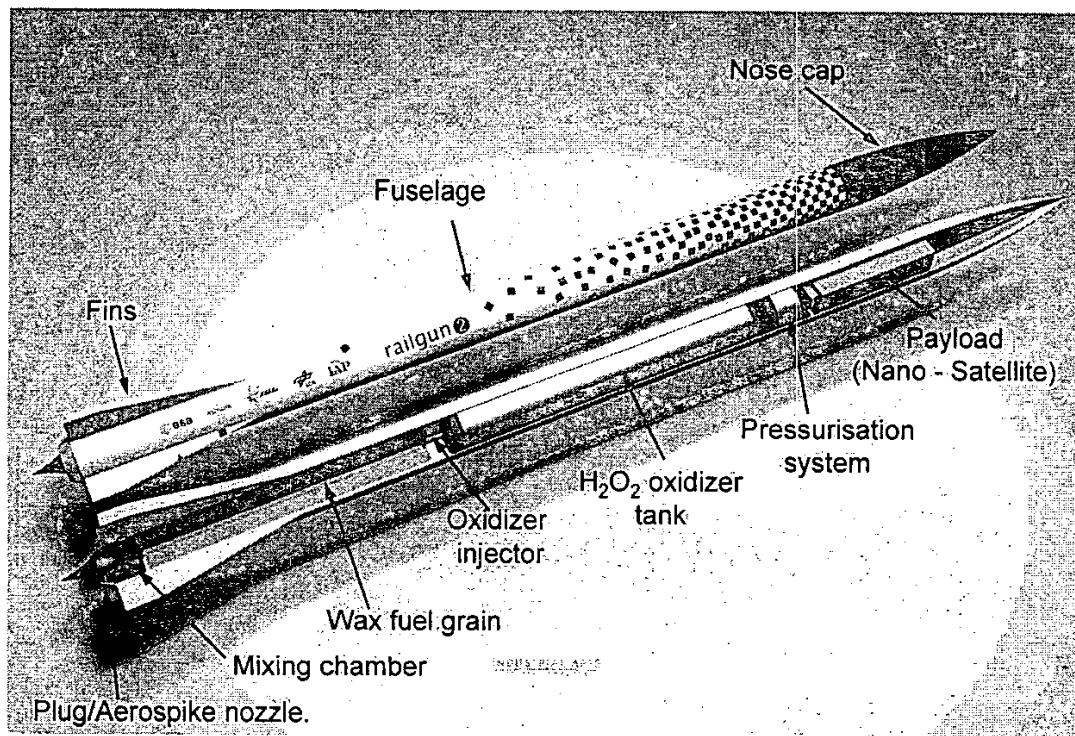


Fig. 15: Layout of projectile with hybrid rocket engine – proposed configuration

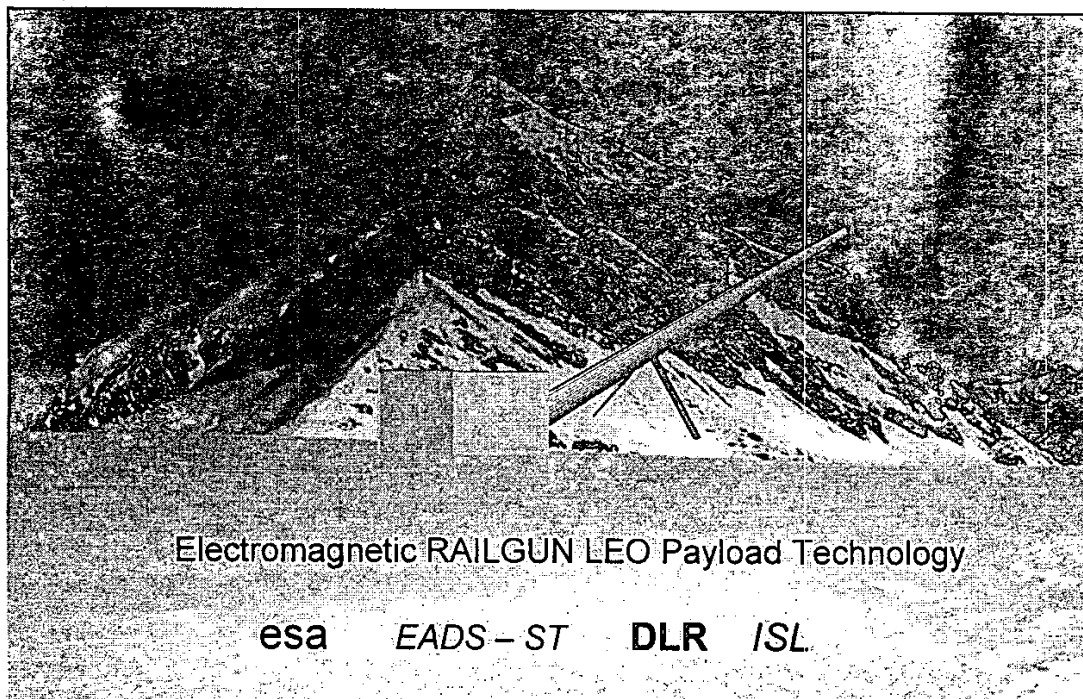


Fig. 16: Electromagnetic RAILGUN LEO Payload launch facility - an artist impression